

Data Dictionary

Definition: → Contains metadata.

⇒ The data dictionary is like the information hub of an Oracle database.

⇒ it is made up of a bunch of tables and views but these are read only.

Means cannot change anything in it, but can read or look all sorts of important data about the database.

What is inside?

Schema Object Definitions:

⇒ These are like the blueprints for everything in the database, like tables, views, indexes and more.

Space Usage:

⇒ Tells us how much space different parts of the database taking up.

Default values:

⇒ If there is a column without a value, the data dictionary might know what to put there.

Integrity Constraints

⇒ These are rules that keep data clean and organized.

User info

⇒ it keeps track of who is using the database and what they are allowed to do.

Auditing

⇒ This is like keeping a log of who's been poking around in the database.

General info

⇒ Anything else important about the database lives/keep here.

Why it's matter?

⇒ Everyone who deals with the database, from end users to application designers and database administrators relies on the data dictionary.

⇒ it helps them understand how the database is set up and what is going on inside it.

⇒ Because it is read-only, everyone can trust that the info in there is accurate.

How the data dictionary is used?

Info

⇒ The data dictionary has three primary uses:

1. Information hub for Oracle:

⇒ Oracle itself relies on the data dictionary to get all sorts of info about users, different parts of the database and storage structures.

⇒ Whenever Oracle needs to know something, it checks the data dictionary.

2. Keeping things updated:

⇒ Whenever someone makes a change to the database, like creating a new table or adding a column, Oracle updates the data dictionary to keep track of it. Any DDL statement is issued.

3. Reference Guide for every user:

⇒ Any Oracle user can use the data dictionary to learn about the database.

⇒ it is like a giant reference book that's always open but you can only read from it.

Purpose of Data dictionary:

⇒ The main purpose of the DD is to store data that describes the structure of the objects in the Oracle database.

⇒ Because of this purpose, there are many views in Oracle data dictionary that provide information about the attributes and composition of the data structures within the database.

Oracle Data Dictionary views:

⇒ To make it easy for users to find the info they need, there are special views in the data dictionary.

⇒ All of the views are identified by their prefixes:

(I) USER View:

What it includes:

- o only the objects that belong to the user's own database schema.

Who can use it?

- o Any user can access this view to see their own objects.

Why it's useful?

- o it is great for users to quickly see and manage their own stuff without being distracted by objects from other users.

(II) DBA View:

What it includes:

- All the objects in the entire database.

Who can use it:

- Only users with DBA privileges can access this view.

Why it's useful:

- It gives administrators a complete overview of everything in the database, making it handy for managing and maintaining the database as a whole.

III) ALL-Views:

What it includes:

- All the objects in the database that the user has access to include their own and objects they have been granted access to in other schemas.

Who can use it:

- Any user can access this view.

Why it's useful:

- ⇒ It provides a comprehensive view of all objects that a user can interact with, making it helpful for users who need to work with objects from multiple schemas.

Example: Tables

DBA TABLES:

- Shows all tables in the entire database, but only accessible to DBAs.

USER TABLES:

- Displays tables owned by the user, for easy management.

ALL TABLES:

- Offers a view of all tables of the user can access, whether they own them or not, simplifying cross-schema data access.

Structure of the data dictionary

⇒ The data dictionary consist of the following:

Base Tables:

⇒ The underlying tables that store information about the associated database.

⇒ Only oracle itself should write and read these tables.

⇒ Users rarely access them directly because they are normalized, and most of the data is stored in a cryptic format.

User-Accesible Views:

⇒ The views that summarize and display the information stored in the base tables of the data dictionary.

⇒ These views decode the base table data into useful information, such as user or table names, using joins and WHERE clauses to simplify the information.

⇒ Most users are given access to the views rather than the base tables.

SYS: keeper of the data dictionary:

⇒ SYS is like the guardian of the data dictionary.

⇒ It owns all the base tables and user-accessible views.

⇒ NO oracle user should ever alter (UPDATE, DELETE, or INSERT) any rows or schema objects contained in the SYS schemas because such an activity compromise data integrity.

⇒ The security administrator must keep strict control of this central account.

Caching the data Dictionary for Quick Access:

Why it's cached:

⇒ Oracle stores much of the data dictionary info in the system Global Area (SGA) to quickly

access it.

⇒ This is important because Oracle frequently checks the data dictionary during database operations to confirm access and check schema objects.

Storage Methods:

⇒ All this information is kept in memory using a smart algorithm called the LRU.

Passing info:

⇒ Details related to passing are usually kept in these caches to further enhance performance.

Comments Columns:

⇒ The COMMENTS columns describing the tables and their columns are not cached unless they are accessed frequently.

Accessibility & Availability:

The data dictionary is readily available whenever the database is operational, residing within the

SYSTEM tablespace, which remains online at all times.